

Katie Knox

Urban planning and spatial analysis · Philadelphia

EDUCATION

Master of City Planning + Master of Urban Spatial Analytics

2025 – 2027

University of Pennsylvania

Dual degree, in progress. Smart City concentration, focused on urban spatial analysis.

Bachelor of Arts

2017 – 2022

Swarthmore College

Sigma Xi, The Scientific Research Honor Society.

EXPERIENCE

Research Assistant

2026 – present

Allison Lassiter Lab, University of Pennsylvania

- Create agent-based models to simulate adaptations to salt-water intrusion in coastal water systems.

Data and Research Manager

2022 – 2026

National Digital Inclusion Alliance

- Mixed-method programme evaluation for national digital equity initiatives, including the National Digital Navigator Corps — survey design, focus groups, statistical analysis and data visualisation.
- Managed and standardised multi-site data collection across 18 organisations, including Tribal governments and rural-serving nonprofits, developing reporting frameworks adaptable across institutional and geographic contexts.

- Original research on digital inequities affecting people of colour with disabilities, using federal datasets and primary data collection.
- Translated spatial and administrative datasets into public-facing research products — StoryMaps, reports — for policymakers and practitioners.
- Presented findings at national webinars and conferences on digital inclusion and public data use.

Web Development Fellow

2020 – 2021

Penn Hill Group

- Built the State Digital Equity Scorecard in React and Node.js for Microsoft Corporate, External and Legal Affairs, to help state leaders meet new Infrastructure Investment and Jobs Act requirements.

Research Assistant

Summer 2020

Gerrymandering Research Lab, Swarthmore College

- Published voting district election data on the open-source mapping tool Districtr.
- Co-authored research on political gerrymandering in Maine and New Hampshire.

Student Developer

Autumn 2018 –
Autumn 2019

Swarthmore College

- Developed Swarthmore College's Civic Engagement Map, the Bach Chorale Workshop, and the 1847 Philadelphia African-American Census data project.

SKILLS

PROGRAMMING AND ANALYSIS

R, Python (pandas), CUDA, spatial statistics, machine learning, remote sensing, deep learning

GIS AND SPATIAL

ArcGIS, QGIS

WEB DEVELOPMENT

React, Node.js, JavaScript

METHODS

Programme evaluation, survey design, mixed-method research

TOOLS

Git, Excel, PowerPoint, Canva, Tableau, Adobe Illustrator and InDesign

LANGUAGES

English (fluent), French (B2)

PAPERS AND PRESENTATIONS

- Asgari, S., Basewitz, Q., Bergmann, E., Brogsol, J., Cox, N., Davis, D., Kampel, M., Keating, B., Knox, K., Lam, A., Lopez-Nava, J., Paige, J., Pitock, N., Song, V., and Torrance, D. (2020). Assessing congressional districting in Maine and New Hampshire. [arXiv preprint arXiv:2011.06555](#)
- Ferguson, T., Huang, A., Knox, K., and Lim, S. Parallelizing the RNA Secondary Structure Dynamic Programming Problem. [Poster presentation, Consortium for Computing in Colleges Northeast Region, 2022](#)